

HOMEWORK 4
due Sat, October 3, 11:59 PM
(Upload your solutions to Gradescope)

1. (5 points) (#1 p. 83.) Expand the function $f(x) = x^2$ in a Fourier series valid on the interval $-\pi \leq x \leq \pi$. Use a computer to plot both f and the partial sums of its Fourier series,

$$S_N(x) = a_0 + \sum_{k=1}^N \left(a_k \cos(kx) + b_k \sin(kx) \right)$$

for $N = 1, 2, 5, 7$. Observe how the graphs of the partial sums $S_N(x)$ approximate the graph of f . Plot the same graphs over the interval $-2\pi \leq x \leq 2\pi$.

2. (4 points) (#8 p. 84.) Expand the function

$$f(x) = \begin{cases} 1 & -1/2 < x \leq 1/2, \\ 0 & -1 < x \leq -1/2 \text{ or } 1/2 < x \leq 1, \end{cases}$$

in a Fourier series valid on the interval $-1 \leq x \leq 1$. Use a computer to plot the various partial Fourier series along with the graph of f as in exercise 1 for $N = 5, 10, 20$ and 40 terms. Notice how much slower the series converges to f in this example than in exercise 1. What accounts for the slow rate of convergence in this example?

3. (6 points) (#20 p. 85.) If $f(x)$ is continuous on the interval $0 \leq x \leq a$, show that its even periodic extension is continuous everywhere. Does this statement hold for the odd periodic extension? What additional condition(s) is (are) necessary to ensure that the odd periodic extension is everywhere continuous?
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4. (2 points) (#19 p. 85.) Let $F(x)$ be the (pointwise sum of the infinite) Fourier series for the function

$$f(x) = \begin{cases} 1 & -1/2 < x \leq 1/2, \\ 0 & -1 < x \leq -1/2 \text{ or } 1/2 < x \leq 1. \end{cases}$$

State the precise value of $F(x)$ for each x in the interval $-1 \leq x \leq 1$.

5. (3 points) (#22 p. 85.) In exercise 1, you found the Fourier series for the function $f(x) = x^2$, $-\pi \leq x \leq \pi$. Explain why this series converges uniformly to x^2 on $[-\pi, \pi]$. Use this to show that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$. (*Hint:* What happens at $x = \pi$?)

Page numbers refer to “A First Course in Wavelets with Fourier Analysis, 2nd Edition” by Boggess and Narcowich.